

# RUIZHI DENG

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## EDUCATION

**Shanghai Jiao Tong University**<sup>†</sup>, Shanghai  
Bachelor of Science in Mechanical Engineering  
GPA: 3.72 / 4.00

Awarded The UZ Scholarship (SJTU Global College), Undergraduate Excellence Scholarship (SJTU)

<sup>†</sup> Dual Degree Program  
Fall 2023- Summer 2027 (Expected)

**University of Michigan**<sup>†</sup>, Ann Arbor, Michigan  
Bachelor of Science in Computer Engineering  
GPA: 4.0 / 4.0

Fall 2025- Spring 2027 (Expected)

**Relevant Courses:** Advanced Embedded Systems, Machine Learning, Computer Vision, Embedded Control Systems, Computer Organization, Fluid Mechanics, Mechanical Behavior of Materials.

## RESEARCH INTERESTS

Robot Learning; Control; Embedded Systems.

## RESEARCH & ENGINEERING SKILLS

- Proficient in C++, C, Python, ROS, Git, CMake.
- Skilled in embedded system development and CAD (STM32, Arduino, and SolidWorks).
- Experience with Video Editing and Photography.
- GitHub: [Ruizhi-Deng \(Ruizhi Deng\)](#), LinkedIn: [Ruizhi Deng | LinkedIn](#)

## PROFESSIONAL EXPERIENCE

**Research Assistant**, HDR Lab, University of Michigan, Ann Arbor

Mar 2026 – Present

- Designed a cable-driven soft manipulator capable of pick-and-place tasks and trajectory following under specified payload conditions using SolidWorks for CAD design and Arduino for low-level control.
- Designed novel control methods for soft manipulators using residual networks and receding horizon control to address the sim-to-real gap in soft manipulator modeling and control; achieved control accuracy of the inverse kinematics for the manipulator's tip within 5 mm.
- Latest work "Residual-Corrected Implicit Mechanics for Actionable Digital Twins of Cable-Driven Soft Robots" accepted by IROS 2026 Late Breaking Results.
- Advised by Prof. [Xiaonan Huang](#)

**Research Assistant**, RAP-Lab, Shanghai Jiao Tong University

Nov 2025 – Present

- Co-proposed a novel path-planning method enabling efficient robotic exploration of unknown environments; implemented algorithms using the ROS2 framework and reduced exploration completion time by up to 12% compared with current SOTA.
- Conducted real-world experiments, including using a four-wheel vehicle to explore complex indoor and outdoor environments.
- Proposed a new online fine-tuning method that enables continuous improvement of the robot's exploration efficiency.
- Latest work "RPV\*: Routing with Probabilistic Vertices and Its Use in Autonomous Exploration" accepted by ISRR 2026.
- Advised by Prof. [Zhongqiang Ren](#)

**Programmer**, SJTU VEX Team, Shanghai Jiao Tong University

Jun 2024 - Aug 2025

- Improved the robustness of the encoder odometer driver program written in C++.
- Enhanced the localization and driving programs of the chassis.
- Implemented trajectory tracking algorithms combining dynamic trajectory generation and Pure Pursuit.
- Refactored legacy codebase to enhance system maintainability and scalability for future iterations.

## ACADEMIC PROJECTS

**Rock-Paper-Scissors Robot**, Introduction to Embedded Systems, University of Michigan

Mar 2026 – Apr 2026

- Developed an embedded system using STM32 that integrated multiple devices, including a camera and LCD screen, to enable human-robot interaction through a rock-paper-scissors game.
- Implemented game logic and integrated software modules across system components to achieve functional operation; achieved a winning rate near 100%.

**The Climber Rover**, Design and Manufacture III, Shanghai Jiao Tong University

May 2025 – Aug 2025

- Designed the mechanical structure of a robot capable of climbing poles with varying diameters and self-locking onto poles without external power.

## HONORS and AWARDS

**Champion & Excellence Award**, The 8th China University Intelligent Robot Creative Competition.

Aug 2025

**Skill Champion**, The 8th China University Intelligent Robot Creative Competition (Preliminary).

Jul 2025

**Division Finalist**, VEX U World Championships.

May 2025

**Skill Champion**, Kechuang Qinghe 2024-2025 National VEX Robotics Elite Competition.

Dec 2024

**Champion**, The 7th China University Intelligent Robot Creative Competition.

Aug 2024